TIPHAINE DELAUNAY

Born in 1997, Paris
Driving license

https://tiphainedelaunay.github.io

EDUCATION

PhD in applied mathematics, *2020 - 2023*

MΞDISIM, Inria & École Polytechnique, Institut Polytechnique de Paris - CNRS, Palaiseau, France

Supervisors : Philippe Moireau and Sébastien Imperiale

Title : *Adaptative observers for wave equations and associated discretization: formulations and analyses*

Defense : December 19 2023, Inria, Palaiseau

Jury :

Laurent Bourgeois	Professor, ENSTA Paris	President
Lucie Baudouin	Research director, CNRS	Rapporteur
Takéo Takahashi	Professor, Université de Lorraine	Rapporteur
Yannick Privat	Professor, Université de Lorraine	Examiner
Sébastien Imperiale	Researcher, Inria	Director
Philippe Moireau	Research director, Inria	Director
Muriel Boulakia	Professor, Univ. Versailles Saint-Quentin	Guest

Engineering diploma *2015 - 2020*

INSA Rouen Normandie, Saint-Etienne du Rouvray, France

Major : Applied mathematics

Master’s degree in applied mathematics *2019 - 2020*

Université de Rouen Normandie, Saint-Etienne du Rouvray, France

Thesis : *3D reconstruction from two-dimensional slices applied to medicine*

RESEARCH EXPERIENCE

Post-doctorate *Since September 2025*

LAGA, Institut Galilée, Université Paris Nord, Villetaneuse, France

Supervisors : Stéphanie Chaillat, Marion Darbas and Jérémy Heleine

Topic : *Reconstruction of the Lamé Parameters of Elastic Waves in an Inclusion*

Post-doctorate *February 2024 - June 2025*

MONC, Inria & Institut de Mathématiques de Bordeaux, Talence, France

Supervisors : Annabelle Collin, Christèle Etchegaray and François Moisan

Topic : *Deciphering tumor response to propranolol in angiosarcomas by mathematical modeling and data assimilation*

Research internship *March - August 2020*

MΞDISIM, Inria & École Polytechnique, Palaiseau, France

Topic : *Data assimilation by observers strategies for wave equations*

Research internship *June - August 2018*

University of Dundee, Dundee, Scotland

Topic : *Elliptic curves and cryptography*

TEACHING EXPERIENCE

Mathematics for engineers *2025*
Sup Galilée, Villetaneuse, France, (Practical class L3) - 39h

Tutored interdisciplinary project, *2024*
Master Cancer Biology of University of Bordeaux, Talence, France, (Project M2) - 5h
Creation and supervision of a project: *Mathematical modeling of the growth of spheroids in free growth and in responses to a non-cytotoxic drug*

Introduction to PDEs and finite differences *2022 & 2023*
ENSTA Paris, Palaiseau, France, (Practical class L3) - 2 x 14h

Finite elements method *2021 & 2022*
ENSTA Paris, Palaiseau, France, (Practical class M1) - 2 x 14h

COMMUNICATIONS

Talks at national and international conferences

- **PICOF** *(Invited)* *2025*
  Hammamet, Tunisia, URL
- **12ème Biennale de la Société des Mathématiques Appliquées et Industrielles** *(Invited)* *2025*
  Carcans-Maubuisson, France, URL
- **Rencontres Normandes sur les apsects théorique et numérique des EDP** *(Invited)* *2024*
  Saint-Etienne du Rouvray, France, URL
- **WAVES** *2022*
  Palaiseau, France, URL
- **ECCOMAS** *2022*
  Oslo, Norvège, URL
- **Rencontre Jeunes Chercheuses, Jeunes Chercheurs : Ondes** *(Invited)* *2022*
  Inria Université Côte d'Azur, Sophia Antipolis, France, URL
- **Rencontre pour les Ondes et leurs applications** *(Invited)* *2022*
  INSA Rouen Normandie, Saint-Etienne du Rouvray, France, URL
- **Congrès des Jeunes Chercheurs en Mathématiques Appliquées** *2021*
  Palaiseau, France, URL

Talk at seminar

- **Seminar MAC upcoming** *(Invited)* *2026*
  Institut de mathématiques de Toulouse, Toulouse, France, URL
- **Seminar Applied mathematics** *(Invited)* *2025*
  Laboratoire de Mathématiques Jean Leray , Nantes Université, Nantes, France, URL
- **Seminar math, bio and images** *(Invited)* *2025*
  LAGA, Université Sorbonne Paris-Nord, Villetaneuse, France, URL
- **Seminar COMMEDIA** *(Invited)* *2025*
  Laboratoire Jacques Louis Lions, Paris, France, URL
- **Seminar EDP de Nancy** *(Invited)* *2024*
  Institut Elie Cartan de Lorraine, Nancy, France
- **Seminar on the mathematical and statistical foundation of future data-driven engineering** *2023*
  Isaac Newton Institute, Cambridge, Royaume-Uni, URL
- **Workshop on assimilation, control and computational speedup** *2023*
  LAGA, Université Sorbonne Paris-Nord, Villetaneuse, France, URL

Posters

- **Journées des rencontres IDEFIX-MΞDISIM-POEMS**, Palaiseau, France *2020 & 2021*

PUBLICATIONS

Adaptative observers for wave equations and associated discretizations: formulations and analysis
T. Delaunay (2023)
Thèse de doctorat, Institut Polytechnique de Paris
https://theses.hal.science/tel-04511683

Uniform boundary stabilization of a high-order finite element space discretization of the 1-d wave equation
T. Delaunay, S. Imperiale, P. Moireau (2024)
Numerische Mathematik
https://inria.hal.science/hal-04172229
DOI: 10.1007/s00211-024-01440-9

Mathematical analysis of an observer for solving inverse source wave problem
T. Delaunay, S. Imperiale, P. Moireau (2025)
Inverse problem and imaging
https://inria.hal.science/hal-04344193
DOI: 10.3934/ipi.2025043

Solving inverse source wave problem - From Carleman estimates to observer design
M. Boulakia, M. de Buhan, T. Delaunay, S. Imperiale, P. Moireau (2025)
Published in *Mathematical Control and Related Fields*
https://hal.science/hal-04788439
DOI: 10.3934/mcrf.2025007

Deciphering tumor response to propranolol in angiosarcomas by mathematical modeling and data assimilation
A. Collin, T. Delaunay, C. Etchegaray, F. Laanani, F. Moisan, L. Pechtimaldjian (2025)
In preparation

SCIENTIFIC POPULARIZATION

Participation to Science fair
Institut Polytechnique de Paris, Palaiseau, France *2021*

Production of a popularization video *2021*
Topic : *Mathematical modelling and computer simulation*
https://www.youtube.com/watch?v=28C1C3UmStE

Animation of "Rendez-vous des jeunes mathématiciennes et informaticiennes" *2021*
Online

Popularization conference in a High School *2021*
Académie de Créteil, online
Topic : *Cryptography*

ACADEMIC INTERESTS

- Inverse problem
- Data assimilation
- Mathematical modelling with PDEs
- Evolution problems
- Wave
- Applications in biology and medicine

PROGRAMMING

- Java
- C++
- Matlab
- Python
- FreeFEM++

SOFTWARES

- LaTeX
- Monolix
- Office
- Photoshop
- Paraview

LANGUAGES

- French: Native
- English: Strong reading, writing and speaking skills (TOEIC 825)
- Spanish: Beginner